

Multilingual Customer Chat Translator & Responder — Requirements Sheet

Owner: [Name] | Date: [Date] | Version: 1.0 | Status: Draft

1. Overview

The Multilingual Customer Chat Translator & Responder is an AI-powered system that detects customer languages, translates messages for support agents, analyzes customer intent and sentiment, and generates suitable responses. It reduces language barriers, improves response speed, and enables consistent multilingual customer support.

2. Preface

Problem: Language barriers can delay customer support and cause misunderstandings.

Solution: Use AI-powered translation, analysis, and response generation to support multilingual conversations.

Goal: Provide fast, accurate, secure, and consistent customer communication across multiple languages.

3. Requirements

3.1 Functional

ID	Requirement
FR1	Auto-detect customer language.
FR2	Translate inbound message into agent language.
FR3	Display original + translation side-by-side.
FR4	Detect intent, sentiment, urgency.
FR5	Generate reply in customer's language using brand tone and policy.
FR6	Provide back-translation for agent verification.
FR7	Allow agent to approve, edit, or reject reply.
FR8	Escalate refunds, legal, safety, angry, low-confidence cases.
FR9	Maintain glossary for product names, legal terms, brand voice.
FR10	Log original, translation, reply, language, confidence, agent action.
FR11	Support auto-responder mode with confidence threshold.

3.2 Non-Functional

- **Latency:** Translation < 2s; suggested reply < 5s.
- **Accuracy:** > 90% meaning preservation; > 95% language detection.
- **Availability:** 99.9% uptime.
- **Security:** TLS, encryption at rest, RBAC, PII redaction.
- **Compliance:** GDPR / CCPA, configurable retention.
- **Scale:** [X] concurrent chats, [Y] languages.

3.3 Integrations

Chat platform, CRM, ticketing, translation API/LLM, SSO, analytics.

3.4 Workflow

Customer message → detect language → translate → analyze → draft reply → agent review → send → log.

3.5 Acceptance Criteria

- Agent reads any supported language in English.
- Customer receives reply in their own language.
- Back-translation available before sending.
- Escalation triggers correctly.
- Audit log complete.
- Latency and accuracy thresholds met.

3.6 Success Metrics

Translation accuracy, first response time, resolution time, CSAT, escalation rate, auto-send accuracy.

3.7 Assumptions & Constraints

- LLM/translation API available.
- Agent language is English.
- Human approval required for MVP.
- No legal certification.

3.8 Approvals

Product: _____ Engineering: _____ Support: _____ Compliance: _____